

Exam.	BE	Regular	
Level	BE	Full Marks	80
Programme	BEI, BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT 704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
 - ✓ Attempt All questions.
 - ✓ The figures in the margin indicate Full Marks.
 - ✓ Assume suitable data if necessary.
1. A discrete time system has input $x(n)$ and output $y(n)$. The input output relation of the system is given by

$$y(n) = \sum_{k=0}^n x(k)$$

Check whether the system is memory less, time invariant and stable or not? [2+2+2]

2. Determine whether the given signal is periodic or not. If the signal is periodic, determine

the fundamental period $x[n] = e^{\frac{j\pi n}{16}} \cos\left\{\frac{n\pi}{17}\right\}$. [5]

3. Define the Region of Convergence (ROC) [1+6]

Find the inverse of $H(z) = (1 + 2z^{-1} + z^{-2}) / (1 - 0.75z^{-1} + 0.125z^{-2})$; ROC $0.25 < |z| < 0.5$

4. Plot the pole-zero on the z -plane and draw Magnitude response (not to the scale) of an LTI system described by the equation, $y(n) = x(n) + 0.8x(n-1) + 0.8x(n-2) + 0.49y(n-2)$. [3+7]

5. Draw the Lattice structure from the following system function.

$$H(z) = \frac{1}{1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}} \quad [10]$$

6. Design the symmetric FIR Low Pass Filter (LPF) for which the desired frequency response is expressed as [10]

$H_d(W) = e^{-jW\tau}$, $|W| \leq W_c$ and 0 elsewhere. The length of the filter should be 7 and $W_c = 1$ rad/sample. Make use of the Hanning window.

7. Kaiser window is to be used to design a linear phase FIR filter that meets following specification [2+2+2]

$$|H(e^{j\omega})| \leq 0.01, \quad 0.21\pi \leq |\omega| \leq \pi$$

$$0.95 \leq |H(e^{j\omega})| \leq 1.05, \quad 0 \leq |\omega| \leq 0.19\pi.$$

Calculate the optimum value of ripple, attenuation and window length.

8. Using Bilinear transformation, design a Butterworth low pass filter which satisfies following conditions:

$$0.9 \leq |H(e^{j\omega})| \leq 1, \quad \text{for } 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(e^{j\omega})| \leq 0.2, \quad \text{for } \frac{3\pi}{4} \leq \omega \leq \pi$$

[12]

Consider sampling frequency of 1 Hz.

9. Compute 8-point DIF-FFT of sequence $x(n) = \{2, 1, 2, 1, 1, 2, 1, 2\}$. [8]

10. Obtain the circular convolution of the following sequences:

$$x_1(n) = \{1, 2, 3, 1\} \text{ and } x_2(n) = \{4, 3, 2, 2\}$$

[6]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2081 Baishakh

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEI, BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT 704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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1. A discrete-time LTI system is given by difference equation $y(n) = x(n) + e^a y(n-1)$. Check this system for BIBO stability. [5]
2. Find the output $y(n)$ of LTI system having impulse response $h(n) = \left(\frac{1}{3}\right)^n u(n-3)$ and input $x(n) = \left(\frac{1}{6}\right)^{n-6} u(n)$. [6]
3. Determine the inverse z-transform of $X(z) = \frac{1}{1-0.8z^{-1}+0.12z^{-2}}$. (i) if ROC is $|z| > 0.6$ (ii) if ROC is $|z| < 0.2$ (iii) if ROC is $0.2 < |z| < 0.6$ [2+2+3]
4. Plot the pole-zero in z plane and draw the magnitude response (not to the scale) of the system described by difference equation $y(n) = 0.67 x(n) - 0.3 x(n-1) + 2.75 y(n-1)$ [3+7]
5. a) Draw the cascaded form structure of $H(z) = 10(1-0.25z^{-1})(1-0.667z^{-1})(1+2z^{-1})/(1-0.75z^{-1})(1-0.125z^{-1})\{1-(0.5+j0.5)z^{-1}\}\{1-(0.5-j0.5)z^{-1}\}$ [5]
b) Draw the lattice structure for the given FIR filter and also check whether the system is stable. $H(z) = 1 + (13/24)z^{-1} + (5/8)z^{-2} + (1/3)z^{-3}$ [5]
6. Design a linear phase FIR filter using suitable window to meet following specifications:

$$0.99 \leq |H(e^{jw})| \leq 1.01, \quad \text{for } 0 \leq |w| \leq 0.3\pi$$

$$|H(e^{jw})| \leq 0.01, \quad \text{for } 0.35\pi \leq |w| \leq \pi$$
 [10]
7. What is Gibbs phenomenon and how can it be minimized? Why Kaiser window is better than other fixed windows in FIR filter design? [3+3]
8. Differentiate between bilinear transformation and impulse invariance. Design a Butterworth digital IIR lowpass filter using bilinear transformation by taking $T = 0.1$ second, to satisfy the following specifications. [2+10]

$$0.6 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.35\pi$$

$$|H(e^{j\omega})| \leq 0.1, \quad 0.7\pi \leq \omega \leq \pi$$
9. Why Decimation in Time Fast Fourier Transform (DITFFT) Algorithm is better than direct computation of DFT? Find 4 point DFT of the sequence $x(n) = \{2, 2, 4\}$ using DITFFT algorithm. [2+6]
10. Compute circular convolution of the following two sequences using DFT. $x(n) = \{1, 2, 4, 5\}$ and $h(n) = \{2, 1, 6, 8\}$ [6]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2080 Baishakh

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Subject: - Digital Signal Analysis and Processing (CT 704)

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1. Check whether following signals are periodic or not. If yes, state their periodic time. [2+2]
 - a) $x[n] = \sin(n\pi) + \cos(n\pi)$
 - b) $x[n] = \sin(3n\pi/5) + \cos(4n\pi/7)$
2. Find the output of LTI system having impulse response $h[n] = (1/2)^n u[n]$ and input $x[n] = 5e^{jn\pi/3}$ for $-\infty < n < \infty$. [5]
3. Define ROC. Find inverse z-transform of $X(z) = (1+2z^{-1}+z^{-2})/(1-1.5z^{-1}+0.5z^{-2})$, ROC: $|z| > 1$. [1+5]
4. Differentiate between FIR system and IIR System. The poles of a system are located at $0.45 \pm j1.6$ and zeros at $0.58 \pm j2.06$. Map the poles and zeros in the z-plane and plot the magnitude response (not in scale) of the system. [4+6]
5. Compute Lattice-ladder coefficients and draw lattice structure for given system [6]

$$H(z) = (2 - 0.7z^{-1} + 0.5z^{-2}) / (1 - 0.3z^{-1} + 0.25z^{-2})$$
6. Realize the given system in Cascade Form of 2nd order section flow graph representation. [4]

$$H(z) = \{(1-0.4z^{-1})(1+0.2z^{-1})(1-0.3e^{j\pi/6}z^{-1})(1-0.3e^{-j\pi/6}z^{-1})\} / \{(1-0.5e^{j\pi/3}z^{-1})(1-0.5e^{-j\pi/3}z^{-1})(1+0.7e^{j\pi/4}z^{-1})(1+0.7e^{-j\pi/4}z^{-1})\}$$
7. In which case do we choose FIR filter and IIR filter? Design a linear phase FIR filter using Kaiser Window to meet the following specifications: [2+8]

$$0.99 \leq |H(e^{jw})| \leq 1.01 \quad \text{for } 0 \leq w \leq 0.016\pi$$

$$|H(e^{jw})| \leq 0.01 \quad \text{for } 0.08\pi \leq w \leq 2\pi$$
8. Explain in detail about how Gibb's oscillation arise while using the rectangular window in FIR filter design. [5]
9. Design a low pass digital IIR filter by Bilinear Transformation method to an approximate Butterworth low pass filter, if passband edge frequency is 0.26π radians and maximum deviation of 0.99 dB below 0 dB gain in the passband. The maximum gain of -14.99 dB and frequency is 0.58π radians in stopband, Consider sampling frequency 0.5 Hz. [11]
10. Describe digital domain Spectral Transformation features and parameters for low pass to high pass in IIR Filter design. [4]
11. How fast is FFT? Find 8-point DFT of sequence $x[n] = \{1, 1, 0, 0, 1, 1, 2\}$ using Decimation in Time Fast Fourier Transform(DITFFT) algorithm. [2+6]
12. Write the complexity of DFT and FFT? Obtain the circular convolution of the following sequences: [2+5]

$$X_1[n] = \{1, 2, 3, 1\} \text{ and}$$

$$X_2[n] = \{4, 3, 2, 2\}$$

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2079 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEI, BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis And Processing (CT 704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define energy and power signal. Determine whether the signal $x[n] = \cos\left[\frac{2\pi n}{5}\right] + \sin\left[\frac{\pi n}{3}\right]$ is periodic or non-periodic and if it is periodic, find its fundamental period. [2+2]
2. Find the output of LTI system having input signal $x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3]$ and $h[n] = 2\delta[n+1] + 2\delta[n-1]$. [5]
3. Find inverses Z-transform of $X(z) = (2z^4 + 2z^3 - 3z + 2)/(z^2 - 1.5z - 1)$, ROC: $|z| < 0.5$, using partial fraction expansion method. [6]
4. Plot the pole-zero in z-plane and draw the magnitude response (not to the scale) of the equation of the system describe by difference equation:
 $y[n] - 0.35y[n-1] + 0.25y[n-2] = x[n] - 0.75x[n-1]$. [3+7]
5. Draw direct form I and Direct form II realization of the following system.
 $y[n] = 0.25y[n-2] + x[n] + 0.4x[n-1] + 0.5x[n-2]$ [2+2]
6. Given a 3-stage lattice filter for all zero polynomial with coefficients $K_1 = 1/4$, $K_2 = 1/2$ and $K_3 = 1/3$. Obtain the system function and FIR filter coefficients of this filter. [6]
7. Define Gibb's phenomenon. Design the FIR filter using Kaiser window technique for the specifications: [2+8]

$$\begin{aligned} 0.899 \leq |H(e^{jw})| &\leq 1 & \text{for } |w| \leq 0.2\pi \\ |H(e^{jw})| &\leq 0.01 & \text{for } 0.4\pi \leq w \leq \pi \end{aligned}$$

8. Discuss the Remez exchange algorithm for FIR filter design. [5]
9. Design a low pass discrete time Butterworth filter using bilinear transformation having following specifications: [11+4]

Passband frequency (W_p) = 0.25π radians
 Stopband frequency (W_s) = 0.55π radians
 Passband ripple (δ_p) = 0.11
 Stopband ripple (δ_s) = 0.21. Consider sampling frequency of 0.5 Hz.

Also, covert the obtained digital low pass filter to high pass filter with new pass band frequency, $W'_p = 0.45\pi$ using digital domain transformation.
10. Why we need FFT? Find the 8-point DFT of the following sequence using radix-2 DITFFT algorithm. [2+6]
11. If $X_1(k)$ and $X_2(k)$ are DFT of sequence $x_1[n] = \{1, 0, 0, 1\}$ and $x_2[n] = \{2, 0, 2\}$ respectively then find the sequence $X_3[n]$; if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k) \cdot X_2(k)$. [7]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2079 Baishakh

Exam.	Back	
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Subject: - Digital Signal Analysis and Processing (CT 704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
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1. Compare between energy signal and power signal. Determine whether the signal

$$x[n] = e^{j(\frac{\pi}{2}n + \frac{4\pi}{7})} \text{ is energy signal or power signal.} \quad [2+2]$$

2. Find the output of LTI system having impulse response $h[n] = \left(\frac{1}{2}\right)^n \{u[n+2] - u[n-2]\}$ to the input $x[n] = \{2, 1, 0, -1, 4\}$. [5]

3. Define z-transform for a discrete time signal. Find the inverse z-transform for $H(z) = \frac{z}{3z^2 - 4z + 1}$ using partial fraction method for $\frac{1}{3} < |z| < 1$. [1+5]

4. Plot the pole-zero in z-plane and draw magnitude response (not to the scale) of the system described by difference equation [3+7]

$$y[n] - 0.3y[n-1] + 0.2y[n-2] = x[n] - 0.5x[n-1]$$

5. Compute Lattice-ladder coefficients and draw lattice structure for given system $H(z) = (1 - 0.4z^{-1} + 0.25z^{-2}) / (1 - 0.3z^{-1} + 0.5z^{-2})$. Also check the stability of given system. [6+1]

6. Obtain the Direct Form I and Direct Form II realization of the following system: [4]

$$y[n] - 0.75y[n-1] - 0.25y[n-2] = x[n] + 0.5x[n-1]$$

7. Design a low pass digital FIR filter having Pass band edge frequency $\omega_p = 0.2\pi$, Stop band edge frequency $\omega_s = 0.45\pi$ and Stop band attenuation $\alpha_s = 51$ dB using any appropriate window function. [5+3]

8. What do you understand by optimum filter? Describe Remez exchange algorithm for FIR filter design along with the flowchart. [1+6]

9. Design a low pass digital IIR filter by Bilinear Transformation method to an approximate Butterworth low pass filter, if passband edge frequency is 0.24π radians and maximum deviation of 0.98 dB below 0 dB gain in the passband. The maximum gain of -14.95 dB and frequency is 0.57π radians in stopband, consider sampling frequency 0.5 Hz. Compare impulse invariance method with bilinear transformation method. [11+3]

10. Why we need DFT? Find 8-point DFT of sequence $x[n] = \{1, 2, 4, 3, 5, -1, 3\}$ using Decimation in Frequency Fast Fourier Transform (DIFFFT) algorithm. [2+6]

11. Find the circular convolution of the sequences $x_1[n] = \{1, -1, -2, 3, -1\}$ and $x_2[n] = \{1, 2, 3\}$. [7]

2078/05/29

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2078 Bhadra

Exam.	Regular		
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Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT 704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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1. Determine whether the signal $x[n] = \cos\left[\frac{\pi n}{2}\right] \cdot \cos\left[\frac{\pi n}{4}\right]$ is periodic or non periodic and if it is periodic, find its fundamental period. [4]
2. Find the output of LTI system having impulse response $h[n] = u[n] - u[n-4]$ and input signal $x[n] = (1/2)^n u[n]$. [5]
3. Define ROC. Find inverse z-transform of $X(z) = (z^3 + z^2 + 1.5z + 0.5)/(z^3 + 1.5z^2 + 0.5z)$, ROC: $|z| < 1/2$. [1+5]
4. Determine the zero-input response for a second order system given by: [4]

$$y[n] - 3y[n-1] - 4y[n-2] = x[n]$$
5. Plot the pole-zero in z-plane and draw magnitude response (not to the scale) of the system described by difference equation. [2+4]

$$y[n] - 0.4y[n-1] + 0.25y[n-2] = x[n] - 0.4x[n-1]$$
6. The system function of a filter is $H(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$. Draw the Direct Form and Lattice Structure implementation of the above filter. [3+7]
7. Design a linear phase FIR filter using KAISER window to meet the following specifications: [8]

$$\begin{cases} |H(e^{jw})| \leq 0.01; & 0 \leq w \leq 0.25\pi \\ 0.95 \leq |H(e^{jw})| \leq 1.05; & 0.35\pi \leq w \leq 0.6\pi \\ |H(e^{jw})| \leq 0.01; & 0.65\pi \leq w \leq \pi \end{cases}$$
8. What is optimum filter? Show mathematical expression of Remez exchange algorithm for FIR filter design. [1+6]
9. Design a LPF Butterworth filter using Impulse Invariance Method (IIM) method with passband and stopband frequencies 200Hz and 500Hz respectively. The passband and stopband attenuations are 5dB and 12dB respectively. The sampling frequency is 5000Hz. What is pre-warping and why it is necessary? Explain. [12+3]
10. Differentiate between DFT and DTFT. Find the circular convolution of $x_1[n] = \{2, 1, 2, 1\}$ and $x_2[n] = \{1, 2, 3, 4\}$ [2+6]
11. Find the 8 - point DFT of $x[n] = u[n] - u[n-4]$ using FFT DIT algorithm. [7]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2076 Chaitra

Exam.	Regular		
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Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT 704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- Define even and odd type discrete time signals with suitable example. Plot the signal $x[-2n+3]$ where $x[n] = \{1, 2, 0, -1, -3, -4\}$. [2+3]
- Determine whether the following system are: [5]
 - $y[n] = x[-n]$ is time-invariant or not.
 - $y[n] = x[n^2]$ is linear or not.
- Find the output of LTI system having input signal $x[n] = u[n+1] - u[n-4]$ and impulse response $h[n] = (1/2)^n u[n-1]$. [6]
- Define ROC of z-transform. Find inverse z-transform using partial fraction expansion of $X(z) = (z^4 + 5z^3 - 3z + 4) / (z^2 - 1.5z - 1)$, ROC: $|z| < 0.5$. [2+6]
- Draw the pole-zero in the z-plane for a system with poles at $0.45 \pm j1.06$ and zeroes at $0.58 \pm j2.06$. Also plot the magnitude response (not to the scale) of the system. [2+6]
- Compute Lattice and Ladder coefficients and Draw lattice-ladder structure for given IIR system $H(z) = (0.5 - 2z^{-1} + 3z^{-2}) / (1 - 0.5z^{-1} - 0.7z^{-2} + 0.3z^{-3})$. [6+4]
- Realize the given system in Cascade form of 2nd order section in signal flow graph representation. [4]

$$H(z) = \{(1 - 0.5z^{-1})(1 + 0.35z^{-1})(1 - 0.3e^{j2\pi/5}z^{-1})(1 - 0.3e^{-j2\pi/5}z^{-1})\} / \{(1 - 0.6e^{j\pi/3}z^{-1})(1 - 0.6e^{-j\pi/3}z^{-1})(1 + 0.5e^{j2\pi/7}z^{-1})(1 + 0.5e^{-j2\pi/7}z^{-1})\}$$
- Design the FIR filter using suitable window for the specifications: [6]

$$0.899 \leq |H(e^{j\omega})| \leq 1, \text{ for } |\omega| \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.01, \text{ for } 0.4\pi \leq \omega \leq \pi$$
- What is optimum filter? Show mathematical expression of Remez exchange algorithm for FIR filter design. [1+5]
- Design a digital low pass Butterworth filter by applying bilinear transformation techniques for the given specifications: [10]
 - Passband peak to peak ripple $\leq 1\text{dB}$
 - Passband edge frequency = 1.2KHz
 - Stopband Attenuation $\geq 40\text{dB}$
 - Stopband edge frequency = 2.5 KHz
 - Sample rate = 8KHz
- Find 8-point DFT of sequence $x[n] = \{1, 2, 3, 3, 5, 0, 4, 6\}$ using Decimation in frequency Fast Fourier Transform (DIFFFT) algorithm. [7]
- Find $x_3[n]$ if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k) * X_2(k)$ where $X_1(k)$ and $X_2(k)$ are 4-point DFT of $x_1[n] = \{1, 2, -2\}$ and $x_2[n] = \{1, 2, 3, -1\}$ respectively. [5]

TRIBHUVAN UNIVERSITY
 INSTITUTE OF ENGINEERING
Examination Control Division
 2076 Ashwin

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Subject: - Digital Signal Analysis and Processing (CT 704)

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1. Explain Fourier transform multiplication property for two sequences. Write Dirichlet's conditions for Fourier series. [4+3]

2. Find convolution between two signals $x[n] = 2^n 4[-n]$, $0 < a < 1$ and $h[n] = 4[n]$ [6]

3. State Convolution property of Z-transform. Find inverse Z-transform of $X(z) = z / \{(z - 0.6)(z + 0.5)^2\}$, ROC: $|z| > 0.6$ [3+6]

4. Describe stability and causality characteristics of LTI system in terms of Impulse Response and ROC of its transfer function with suitable examples. [4+3]

5. Compute Lattice and Ladder coefficients and Draw lattice-ladder structure for given IIR system $H(z) = (0.7 - 1.5z^{-1} + 0.5z^{-2}) / (1 - 0.5z^{-1} - 0.7z^{-2} + 0.3z^{-3})$ [6+3]

6. For the system described by the following difference equation: $y[n] = 0.67x[n] - 0.3x[n-1] + 2.75y[n-1]$ [2+8]

Map the poles and zero in the z-plane and plot the phase response of the system.

7. Design a low pass discrete IIR filter by Bilinear Transformation method to an approximate Butterworth filter having specifications as below: [12]

Pass bandedge frequency (ω_p) = 0.22π radians

Stop bandedge frequency (ω_s) = 0.54π radians

Passband ripple (δ_p) = 0.11

Stopband ripple (δ_s) = 0.22, Consider sampling frequency 0.5 Hz.

8. Why we need DFT? Find 8-point DFT of sequence $x[n] = \{1, 2, 3, 3, 5, 1, 4, 2\}$ using Decimation in frequency Fast Fourier Transform (DIFFFT) algorithm. [2+8]

9. In which case do we choose FIR filter and IIR filter? Design a Kaiser Window to meet the following specifications. [2+4+4]

$$0.99 \leq |H(e^{jw})| \leq 1.01, \quad \text{for } 0 \leq w \leq 0.16\pi$$

$$|H(e^{jw})| \leq 0.01, \quad \text{for } 0.18\pi \leq w \leq 2\pi$$

Draw the flow chart for Remez-Exchange algorithm

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2075 Chaitra

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Subject: - Digital Signal Analysis and Processing (CT 704)

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1. Define Power and Energy type discrete time signal with suitable example. Differentiate between Fourier Series and Fourier Transform. [3+4]
2. Find the output of LTI system having impulse response $h[n]$ with $h[-2] = 3$, $h[0] = 2$, $h[1] = 1$ and input signal $x[n] = (2)^n$, for $-1 \leq n \leq 3$. Also check the answer. [5+2]
3. Plot the pole-zero in z-plane and draw magnitude response (not to scale) of the system described by differential equation

$$y(n) - 0.3y(n-1) = 2x(n-2) + 0.7x(n-1) + 4x(n) \quad [2+7]$$

4. Draw the lattice structure from the following system function

$$H(z) = \frac{1}{1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}} \quad [9]$$

5. What is optimum filter? Show mathematical expression of Remez exchange algorithm for FIR filter design. [2+6]
6. List out the properties of Region of convergence and locate the ROC of the following signal

$$x[n] = (0.1)^n u[n] + (0.3)^n u[-n-1] \quad [4+6]$$

7. Using bilinear transformation, design a digital filter using Butterworth approximation which satisfies the following conditions

$$0.8 \leq |H e^{jW}| \leq 1 \text{ for } 0 \leq W \leq 0.2\pi$$

$$|H e^{jW}| \leq 0.2 \text{ for } 0.6\pi \leq W \leq \pi$$

[10]

8. How fast is FFT? Find $X(3)$ and $X(5)$ for given sequence $x[n] = \{1, -2, 3, 2\}$ using DITFFT algorithm. [2+8]

9. Differentiate between linear convolution and circular convolution compute circular convolution of signals

$$X_1[n] = \{0, 0, 1, 1\} \text{ and } X_2[n] = \{1, 1, 1, 1\}$$

[3+7]

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Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Determine whether the following sequences are linear or not:

[3+3]

a) $y[n] = x^2[n]$

b) $y[n] = \cos\left(\frac{5\pi}{8}n + \frac{\pi}{4}\right)$

2. Find the output of LTI system having impulse response $h[n] = 2^n * \{u[n] - u[n-3]\}$ and input signal $x[n] = \delta[n] + \delta[n-1] + \delta[n-2]$.

[5]

3. List out the properties of Region of convergence and locate the ROC of the following signal.

[3+6]

$$x[n] = (0.6)^n u[n] + (0.25)^n u[n]$$

4. Draw the poles and zeros in the z-plane for a system with poles at $0.45 \pm j1.06$ and zeros at $0.58 \pm j2.06$. Also plot the magnitude response of the system.

[2+8]

5. Draw the Lattice structure from the following system function:

[7+3]

$$\frac{1}{3 + \frac{39}{24}Z^{-1} + \frac{15}{8}Z^{-2} + \frac{3}{9}Z^{-3}}$$

And represent $\frac{5}{8}$ and $-\frac{5}{8}$ in sign magnitude, 1's complement and 2's complement format.

[12]

6. Design a digital low-pass filter with the following specification:

- i) Pass-band magnitude constant to 0.7 dB below the frequency of 0.15π
- ii) Stop-band attenuation at least 14 dB for the frequencies between 0.6π to π

Use Butterworth approximation as a prototype and use bilinear transformation method to obtain the digital filter.

7. Design a linear phase FIR filter using Kaiser Window to meet the following specifications:

[8+4]

$$0.99 \leq |H(e^{jw})| \leq 1.01, \quad \text{for } 0 \leq w \leq 0.19\pi$$

$$|H(e^{jw})| \leq 0.01, \quad \text{for } 0.21\pi \leq w \leq \pi$$

Draw the flow chart for Optimum filter design.

8. How fast is FFT compare to DFT? Draw the butterfly diagram of 8-point DFT of a sequence as $x[n] = n+1$ using Decimation in Time FFT algorithm.

[3+7]

9. State the circular convolution property of DFT. Find the circular convolution of:

[1+5]

$$x_1(n) = \{1, 2, -1, 1\} \text{ and } x_2(n) = \{1, 3, 5, 7\}$$

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Plot the sequence $x[n] = u[n] - u[n-3] + 5\delta[n-4] = nu[n-6]$. List out the properties of LTI system. [3+2]
2. Determine whether the following system are: [3+3]
 - a) $y[n] = y[n-4] + x[n-4]$ is Time-invariant or not
 - b) $y[n] = x^2[n]$ is Linear or Non-linear
3. Define a ROC. What are the properties of ROC of z-transform? Find the inverse Z-transform of $X(z) = (2z^2 + 2z^2 + 3z + 5)/(z^2 - 0.1z - 0.2)$, ROC: $|z| < 0.4$. [1+3+5]
4. The poles of a system are located at: $0.45 - 0.77i$ and $-2 \pm 0.3i$. Map the poles and zero in the z-plane and plot the magnitude response of the system. [2+8]
5. Obtain the Direct Form I and Direct Form II realization of the following system. [5]

$$3y[n] + y[n-1] + 2y[n-4] = 2x[n] + x[n-3]$$
6. Determine the lattice coefficients corresponding to the FIR filter with the system function: [5]

$$H(z) = A_3(z) = 1 + \frac{52}{96}z^{-1} + \frac{25}{40}z^{-2} + \frac{1}{3}z^{-3}$$
7. Design a digital low-pass filter with the following specification: [12]
 - i) Pass-band magnitude constant to 0.7 dB below the frequency of 0.15π
 - ii) Stop-band attenuation at least 14 dB for the frequencies between 0.6π to π
 Use Butter worth approximation as a prototype and use impulse invariance method to obtain the digital filter.
8. Design a FIR linear phase filter using Kaiser window that meets the following specifications: [9+3]

$$|H(e^{jw})| \leq 0.01, 0 \leq |w| \leq 0.25\pi$$

$$0.95 \leq |H(e^{jw})| \leq 1.05, 0.35\pi \leq |w| \leq 0.6\pi$$

$$|H(e^{jw})| \leq 0.01, 0.65\pi \leq |w| \leq \pi$$
 Also determine the minimum length $(M+1)$ of the impulse response and Kaiser window parameter β .
9. Why do we need DFT? Draw the butterfly structure to compute the DFT of the following signal using Radix-2 DIFFFT algorithm, and compute $X(2)$ and $X(1)$ only $x[n] = \{1.5, -1, 1.8, 0.6, 3, 1.7\}$ [3+7]
10. Define zero padding. Find the linear convolution through circular convolution with padding of zeros for the following sequences: $x[n] = \{1, 1, 1, 1\}$ and $h[n] = \{2, 3\}$. [1+5]

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define Energy and Power type discrete time signal. Check whether signal $x[n] = e^{j(\pi n/3 + \pi/4)}$ is periodic or not. If it is periodic, state its periodic time. [2+2]
2. Find the output of LTI system having impulse response $h[n] = (1/2)^n \{u[n+2] - u[n-2]\}$ and input signal $x[n] = \{2, 1, 0.5, -1\}$. Also check the answer. [3+2]
3. State and explain the properties of a Region of Convergence (ROC). Find the inverse z-transform of $X(z) = z^2 \left[1 - \frac{3}{2}z^{-1} \right] (1 + z^{-1})(1 - z^{-1})$ [3+3]
4. Plot the pole-zero in z-plane and Draw Magnitude Response (not to the scale) of the system described by difference equation $y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.5x[n-1] + 0.6x[n-2] + 0.8x[n-3]$ [3+7]
5. Draw the direct form and Lattice structure of a filter with system function $H(z) = 1 + 0.7z^{-1} + 1.2z^{-2} - z^{-3}$. [3+7]
6. Why Kaiser window is better than other fixed windows in FIR filter design? Find out first six coefficients of impulse response of a low pass FIR filter having Pass band edge frequency $\omega_p = 0.2\pi$, Stop band edge frequency $\omega_s = 0.5\pi$ and Stop band attenuation $\alpha_s = 41\text{dB}$ using any appropriate window function. [2+6]
7. What is an optimum filter? Show mathematical expression of the Remez exchange algorithm for FIR filter design with flow chart. [1+6]
8. Design a low pass discrete IIR filter by Bilinear Transformation method to an approximate Butterworth filter having specifications as below: [15]

Pass bandedge frequency (ω_p) = 0.27π radians
 Stop bandedge frequency (ω_s) = 0.58π radians
 Passband ripple (δ_p) = 0.11
 Stopband ripple (δ_s) = 0.21, Consider sampling frequency 0.5 Hz.
9. Compute the 8-point DFT of the sequence $x[n] = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0 \right\}$ using Decimation in Frequency Fast Fourier Transform (DIF-FFT) algorithm. [7]
10. What is a zero padding? If $X_1(k)$ and $X_2(k)$ are DFT of sequence $x_1[n] = \{1, 2, 0, 1, -2\}$ and $x_2[n] = \{1, 0, 1, 1, 2\}$ respectively then find the sequence $x_3[n]$; If DFT of $x_3[n]$ is given by $X_3(k) = X_1(k), X_2(k)$. [1+7]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Define and plot a discrete time unit step signal. Explain its relation with unit impulse signal. [1+2]
2. Calculate $y[n]$, if $x[n]$ is $x[-2] = 0.5$, $x[0] = 1$, $x[1] = 0.75$, $x[3] = 0.5$ and $n[n]$ is $n[0] = 1$, $n[1] = 0.75$ and $n[2] = 0.5$ and verify your result. [6]
3. Define a ROC. Find inverse Z-transform of $X(z) = (2z^3 + 2z^2 + 3z + 5) / (z^2 - 0.1z - 0.2)$, ROC: $|z| < 0.4$ [1+5]
4. Define the difference equation with example. The Poles of a system are located at: $0.45 + 0.77i$ and $2 \pm 0.7i$ and zeros at: $1.2 \pm 0.43i$. Plot the magnitude response of this system. [2+8]
5. Draw the Lattice Structure from the following system function: [10]

$$\frac{1 + \frac{1}{3}z^{-1} + \frac{9}{8}z^{-2} + \frac{4}{3}z^{-3} + z^{-4}}{1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}}$$

6. Design a digital Butterworth low pass filter satisfying the constraints

$$\begin{cases} 0.707 \leq |H(e^{jw})| \leq 1 & 0 \leq w \leq \frac{\pi}{2} \\ |H(e^{jw})| \leq 0.2 & \frac{3\pi}{4} \leq w \leq \pi \end{cases}$$

With $T = 1\text{sec}$ using bilinear transformation method. Realize the filter using the most convenient realization form. [11+4]

7. Design an FIR linear phase filter using Kaiser window to meet the following specifications: [8]

$$0.98 \leq |H(e^{jw})| \leq 1.02, \text{ for } 0 \leq w \leq 0.9\pi$$

$$|H(e^{jw})| \leq 0.01, \text{ for } 0.14\pi \leq w \leq \pi$$

8. Draw the Howchart of Remez-Exchange theorem and explain it. [7]
9. Why we need FFT? Find 8-point DFT of sequence $x[n] = \{1, -1, 3, 2, 1, 1, 3, -2\}$ using Decimation in frequency Fast Fourier Transform (DIFFFT) algorithm. [2+6]
10. Find $x_3[n]$ if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k) X_2(k)$ where $X_1(k)$ and $X_2(k)$ are 5-point DFT of $x_1[n] = \{1, -2, 5, 1, 2\}$ and $x_2[n] = \{1, 2, -3, -2\}$ respectively. [7]

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Explain the process of calculating fourier series coefficients. [3]
2. Determine the system output $y(n)$ of the following signals: $h(n) = \{1,1\}$ and $x(n) = \{1,1,1,1\}$ [6]
3. Define a ROC. Find inverse Z-transform of $X(z) = z / \{(z - 0.4)(z + 1.5)^2\}$, ROC: $|z| < 0.4$ [1+5]
4. State linear constant coefficient difference equation and corresponding system function.
Determine the output sequence of the system with impulse response $h[n] = (1/2)^n u[n]$ when the input signal is $x[n] = 10 - 5 \sin(\pi n / 2) + 20 \cos \pi n$ $-\infty < n < \infty$. [3+7]
5. The system function of a filter is $H(z) = 2 + 1.8z^{-1} - 1.6z^{-2} + z^{-3}$. Draw the Direct Form and Lattice Structure implementation of the above filter. [3+7]
6. Explain in detail about how rectangular window is used in FIR filter design. How Gibb's oscillations arise in this process. [6]
7. Explain about Remaz exchange algorithm with suitable derivation and flow chart. [9]
8. Using bilinear transformation, design a butterworth low pass filter which satisfies the following Magnitude Response. [12]

$$0.89125 \leq |H(e^{j\omega})| \leq 1 \quad \text{for } 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.17783 \quad \text{for } 0.3\pi \leq \omega \leq \pi$$
9. Explain briefly about bilinear transformation method of IIR filter design. [3]
10. Why do we need DFT? Find 8-point DFT of sequence $x[n] = \{1, -1, 2, 2, 1, 1, 2, 2\}$ using Fast Fourier Transform algorithm. [2+6]
11. Find $x_3[n]$ if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k) X_2(k)$ where $X_1(k)$ and $X_2(k)$ are 5-point DFT of $x_1[n] = \{1, -2, 2, 1, 4\}$ and $x_2[n] = \{2, 1, -3, -1\}$ respectively. [7]

12/16

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. How fourier series coefficients are calculated? Explain. [4]
2. Find the output of LTI system having impulse response $h[n]$ with $h[-2] = 1$, $h[0] = 2$, $h[1] = 3$ and input signal $x[n]$ with $x[0] = 1/2$, $x[2] = 2$, $x[3] = 3$. Also check the answer. [3+2]
3. Explain the properties of Region of Convergence with examples. [6]
4. Describe stability and causality characteristics of LTI system in terms of Impulse Response and ROC of its transfer function with suitable examples. [4]
5. Plot the pole-zero in z-plane and Draw Magnitude Response (not to the scale) of the system described by difference equation. [2+4]
 $y[n] - 0.4y[n-1] + 0.1y[n-2] = x[n] + 0.6x[n-1]$
6. Determine the Direct Form I and Direct Form II realization of the following system. [5]
 $y(n) = -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-2) + 0.6x(n-2)$
7. Compute the lattice coefficients and draw the lattice structure of following FIR system. [5]
 $H(z) = 1 + 2z^{-1} + z^{-2}$
8. Describe how digital FIR filter can be design by window method. Why Kaiser window is better than other fixed windows in FIR filter design? [5+3]
9. What is an optimum filter? Show mathematical expression of Remez exchange algorithm for FIR filter design. [1+6]
10. Explain about the advantages of selecting bilinear transformation method over impulse invariance method (I I M). Design a digital low pass Butterworth filter using impluse invariant transformation with pass band and stop band frequencies 200Hz and 500Hz respectively. The pass band and stop band attenuation are -5dB and -12dB respectively. The sampling frequency is 5kHz. Use IIM method. [3+12]
11. Find the FFT of the signal $x[n] \{1, 1, 2, 4, 3, 1, 2, 1\}$ using DIT-FFT algorithm. [8]
12. Compute Circular Convolution of $h(n) = \{1, 2, 1, -1, 1\}$ and $x[n] = \{1, 2, 3, 1\}$. [7]

Exam.	New Batch (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define energy and power signal. Check the signal $x[n] = u[n]$ and $x[n] = \delta[n]$ is Energy or Power type. [2+3]
2. Find the output of LTI system having impulse response $h[n] = (1/3)^n \{u[n+1] - u[n-2]\}$ and input signal $x[n] = \{2, 1, 0.5, 3\}$. [5]
3. State the properties of region of convergence (ROC). Drive the convolution property of Z-transform. [3+3]
4. Find the output of LTI System having impulse response $h[n] = (1/2)^n u[n]$ and input signal $x[n] = 5e^{j\pi n/3}$ for $-\infty < n < \infty$. [4]
5. Plot Magnitude Response (not to the scale) of the system described by difference equation. [6]
 $y[n] - 0.5y[n-1] + 0.3y[n-2] = x[n] + 0.7x[n-1]$
6. Determine the Direct Form II realization of the following system [4]
 $y(n) = -0.1y(n-1) + 0.72y(n-2) + 0.7x(n) - 0.252x(n-2)$
7. Compute the lattice coefficients and draw the lattice structure of following FIR system [6]
 $H(z) = 1 + 2z^{-1} - 3z^{-2} + 4z^{-3}$
8. Draw the flowchart of Remez-Exchange theorem and explain it. Design an FIR linear phase filter using Kaiser window to meet the following specifications: [6+8]
 $0.99 \leq |H(e^{jw})| \leq 1.01, \text{ for } 0 \leq w \leq 0.19\pi$
 $|H(e^{jw})| \leq 0.01, \text{ for } 0.21\pi \leq w \leq \pi$
9. Design a low pass digital filter by Bilinear Transformation method to an approximate Butterworth filter, if passband edge frequency is 0.25π radians and maximum deviation of 1 dB below 0 dB gain in the passband. The maximum gain of -15 dB and frequency is 0.45π radians in stopband, Consider sampling frequency 1Hz. [15]
10. Find 8-point DFT of sequence $x[n] = \{1, 1, 0, 1, 0, 1, 2\}$ using Decimation in Time Fast Fourier Transform (DITFFT) algorithm. [7]
11. Why we need DFT? If $X_1(k)$ and $X_2(k)$ are DFT of sequence $x_1[n] = \{1, 2, 4\}$ and $x_2[n] = \{-1, 2, 3, 1\}$ respectively, then find the sequence $x_3[n]$, if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k) X_2(k)$. [2+6]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Find the even and odd part of signal $x[n]$, [3]

$$x[n] = \begin{cases} 1 & \text{for } -4 \leq n \leq 0 \\ 2 & \text{for } 1 \leq n \leq 4 \end{cases}$$

2. A discrete time LTI system has impulse response $h(n) = \{1, 3, 2, -1, 1\}$ for $-1 \leq n \leq 3$. Determine the system output $y(n)$ if the input $x(n)$ is given by $x(n) = 2\delta(n) - \delta(n-1)$. [6]

3. Define ROC. Find inverse Z-transform of [1+5]

$$X(z) = 1/\{(z - 0.5)(z + 2)\}, \text{ if}$$

- i) ROC: $0.5 < |z| < 2$
- ii) ROC: $|z| < 0.5$
- iii) ROC: $|z| > 2$

4. The poles of a system are located at: $0.45 + 0.77i$ and $-2 \pm 0.3i$ and zeroes at: $1.2 \pm 3i$. Map the poles and zero in the z-plane and plot the magnitude response of the system. [2+8]

5. Compute Lattice coefficients and draw lattice structure for given IIR system $H(z) = 1/(1 - 0.01z^{-1} - 0.23z^{-2} + 0.5z^{-3})$. Also check the stability of given system. [4+2+1]

6. What is limit cycle effect in recursive system? Describe with one example showing how it occurs. [3]

7. Design a low pass FIR filter having Pass band edge frequency $\omega_p = 0.3\pi$, Stop band edge frequency $\omega_s = 0.5\pi$ and Stop band attenuation $\alpha_s = 40$ dB using any appropriate window function. [8]

8. What is optimum filter? Show mathematical expression of Remez exchange algorithm for FIR filter design. [1+6]

9. What is the advantage of bilinear transformation? Design a low pass discrete time Butterworth filter applying bilinear transformation having specifications as follows: [2+9+4]

Pass band frequency (ω_p) = 0.25π radians

Stop band frequency (ω_s) = 0.55π radians

Pass band ripple (δ_p) = 0.11

and stop band ripple (δ_s) = 0.21

Consider sampling frequency 0.5 Hz.

Also, convert the obtained digital low-pass filter to high-pass filter with new pass band frequency (ω'_p) = 0.45π using digital domain transformation.

10. Why do we need Discrete Fourier Transform (DFT) although we have Discrete-time Fourier Transform (DTFT)? Find circular convolution between $x[n] = \{1, 2\}$ and $y[n] = u[n] - u[n-4]$. [2+5]

11. How fast is FFT? Draw the butterfly diagram and compute the value of $X(7)$ using 8 pt DIT-FFT for the following sequences: [2+6]

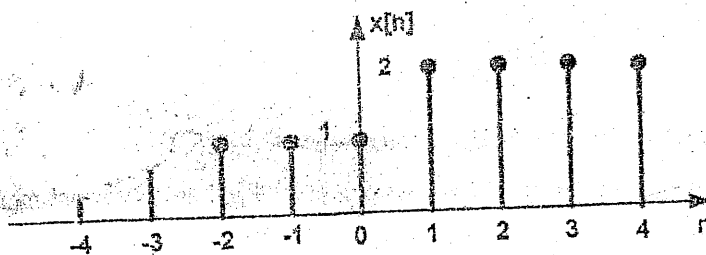
$$x(n) = \{1, 0, 0, 0, 0, 0, 0, 0\}$$

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Find the odd and even part of the following signal:

[4+5]



A discrete time LTI system has input signal and impulse response as,

$$x[n] = \begin{cases} 1 & -1 \leq n \leq 1 \\ 0 & \text{elsewhere} \end{cases} \text{ and } h[n] = \begin{cases} 1 & -1 \leq n \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the output of the system using graphical method.

2. Find the inverse z transform of:

[6]

$$X(Z) = (1+2z^{-1}+z^{-2})/(1+1.5z^{-1}+0.5z^{-2}), |z| > 1$$

using partial fraction method.

3. Why do we need difference equation? State linear constant coefficient difference equation and corresponding system function.

[2+3+5]

Consider an LTI system with impulse response $h[n] = (1/2)^n u[n]$. Determine $y[n]$, if the input is $x[n] = Ae^{int}$

4. If a 3 stage lattice filter for all pole polynomial has coefficients.

[5]

$$K_1 = \frac{1}{4}, K_2 = \frac{1}{2} \text{ and } K_3 = \frac{1}{3} \text{ Obtain the system function of this filter.}$$

5. What is the importance of quantization in Digital Signal Processing? Which one is better rounding or truncation? Explain about limit cycles in recursive system? Define dead band.

[1+1+2+1]

6. Explain in detail about how rectangular window is used in FIR filter design. How Gibb's oscillations arise in this process.

[6]

7. What is a Remez exchange algorithm? Derive its equation and draw its flow chart.

[9]

8. Design a low pass digital filter by Bilinear Transformation method to an approximate Butter worth filter it passband frequency is 0.2π radians and maximum deviation of 1 db below 0 dB gain in the pass band. The maximum gain of -15 db and frequency is 0.4π radians in stop band, consider sampling frequency 1 Hz.

[15]

9. A system has input signal $x[n] = \{1, 2, 3, 4\}$ and impulse response $h[n] = \{1, 3, 5, 7\}$ and the DFT of $x[n]$ is $X[k]$ and the DFT of $h[n]$ is $H[k]$. Find the output of the system $y[n]$ if $G[k] = X[k].H[k]$

[7]

10. Find DFT for $\{1, 1, 2, 0, 1, 2, 0, 1\}$ using FFT DIT butterfly algorithm and plot the spectrum.

[6+2]

Exam.	Regulation		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Determine which of the following signals are periodic and compute their fundamental period: [3]
 - i) $\cos(\pi n^2/8)$
 - ii) $\cos(n/2) \cos(\pi n/4)$
2. Find output, $y(n)$ when: $h(n) = \{5, 4, 3, 2\}$ and $x(n) = \{1, 0, 3, 2\}$ [6]
3. List out the properties of Region of Convergence. Find the Z-transform and locate the ROC of the signal. [2+4]

$$x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[-n-1]$$
4. Find the output of LTI System having impulse response [4]

$$h[n] = (1/3)^n u[n] \text{ and input signal } x[n] = 5e^{j\pi n/2} \text{ for } -\infty < n < \infty.$$
5. Plot Magnitude Response (not to the scale) of the system described by difference equation. $y[n] - 0.3 y[n-1] + 0.225 y[n-2] = x[n] + 0.5 x[n-1]$ [6]
6. Determine the Cascade Form realization of the following system. [4]

$$y[n] - \frac{3}{4} y[n-1] + \frac{1}{8} y[n-2] - x[n] - 2x[n-1] = 0$$
7. Compute the lattice coefficients and draw the lattice structure of following FIR system [6]

$$H(z) = 1 + 3.1z^{-1} + 5.5z^{-2} + 4.2z^{-3} + 2.3z^{-4}$$
8. Describe how FIR filter can be designed by window method. Discuss the characteristics of different type of window function. [4+4]
9. What is an optimum filter? Show mathematical expression of Remez exchange algorithm for FIR filter design. [1+6]
10. Using bilinear transformation method, design a digital filter using Butterworth approximation which satisfies the following conditions: [10]

$$\begin{aligned} 0.8 \leq |H e^{j\omega}| \leq 1 & \quad \text{for } 0 \leq \omega \leq 0.2\pi \\ |H e^{j\omega}| \leq 0.2 & \quad \text{for } 0.6\pi \leq \omega \leq \pi \end{aligned}$$
11. A digital LPF with cut off frequency $\omega_c = 0.2575 \pi$ is given as $H(Z) = \frac{0.1 + 0.4z^{-1}}{1 - 0.6z^{-1} + 0.1z^{-2}}$ [5]

Design a digital high pass filter with $\omega'_c = 0.3567\pi$.
12. Define Padding zones. Find 8-point DFT of sequence. [1+6]

$$x[n] = \{1, 1, 0, 0, 1, 1, 2\}$$
 using Decimation in Time Fast Fourier Transform (DITFFT) algorithm.
13. Why we need DFT? State and prove Circular Convolution property of DFT. [2+2+4]

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Find the even and odd part of signal $x[n]$, [3]

$$x[n] = \begin{cases} 1 & \text{for } -4 \leq n \leq 0 \\ 2 & \text{for } 1 \leq n \leq 4 \end{cases}$$
2. Illustrate the significance of convolution summation in digital signal analysis. Compute the convolution of the following signals: $h(n) = \{1, 0, 1\}$ and $x(n) = \{1, -2, -2, 3, 4\}$ [2+4]
3. Define Region of Convergence. Find inverse Z - transform of $X(z) = z / \{(z-1)(z-2)^2\}$, ROC: $|Z| < 1$ [1+5]
4. Given $H(z)$ for a system with the following difference equation: $y(n) = x(n) + x(n-2)$ [2+6+2]

Plot its poles and zeros in Z plane. Determine its magnitude response. Also, determine whether system is causal and stable.
5. Draw lattice structure for given pole - zero system [6]

$$H(z) = (0.5 + 2z^{-1} + 0.6z^{-2}) / (1 - 0.3z^{-1} + 0.4z^{-2})$$
6. What do you mean by Limit Cycle? How it occurs in recursive system? [1+3]
7. What is the condition satisfied by Linear phase FIR filter? Show that the filter with $h(n) = \{-1, 0, 1\}$ is a linear phase filter. [2+4]
8. Use Hanning window method to design a digital low-pass FIR filter with pass-band edge frequency (ω_p) = 0.25π , stop-band edge frequency (ω_s) = 0.35π where main lobe width of Hanning window is $8\pi/M$, M is the filter length. [9]
9. Why Spectral Transformation is required? [2]
10. Design a low pass digital filter by impulse invariance method to an approximate Butterworth filter, if passband edge frequency is 0.2π radians and maximum deviation of 0.5 dB below 0 dB gain in the passband. The maximum gain of -15 dB and frequency is 0.35π radian in stopband, consider sampling frequency 1Hz. [13]
11. Why do we need Discrete Fourier Transform (DFT) although we have Discrete-time Fourier Transform (DTFT)? Find circular convolution between $x[n] = \{1, 2\}$ and $y[n] = u[n] - u[n-4]$. [2+5]
12. How fast is FFT? Draw the butterfly diagram and compute the value of $x(7)$ using 8 pt DIT-FFT for the following sequences: [2+6]

$$x(n) = \{1, 0, 0, 0, 0, 0, 0, 0\}$$

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Digital Signal Analysis and Processing (CT704)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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1. Define Energy and Power type signal with suitable example. Check the signal $x[n] = \cos(2n\pi/5) + \sin(\pi n/3)$ is periodic or not. [2+2]
2. Define LTI system. Find the output of LTI system having impulse response $h[n] = 2u[n] - 2u[n-4]$ and input signal $x[n] = (1/3)^n u[n]$. [1+4]
3. State the properties of region of convergence (ROC)? Derive the time shifting property of Z-transform. [3+3]
4. Why do we need Difference Equation? Draw Pole-zero in Z-Plane and plot magnitude response (not to the scale) of the system described by difference equation $y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.1x[n-1] - 0.06x[n-2]$. [2+2+6]
5. Determine the Direct Form II realization of the following system $y(n) = -0.1y(n-1) + 0.72y(n-2) + 0.7x(n) - 0.252x(n-2)$. [4]
6. Compute the lattice coefficients and draw the lattice structure of following FIR system $H(z) = 1 + 2z^{-1} - 3z^{-2} + 4z^{-3}$. [6]
7. Design a digital FIR filter for the design of the low pass filter having $\omega_p = 0.3\pi$, $\omega_s = 0.5\pi$, $\alpha_s = 40$ dB using suitable window function. [8]
8. What is optimum filter? Describe Remez exchange algorithm for FIR filter design with flow chart. [1+6]
9. What is the advantage of bilinear transformation? Design a low pass discrete time Butterworth filter applying bilinear transformation having specifications as follows: [2+9+4]
 - Pass band frequency (ω_p) = 0.25π radians
 - Stop band frequency (ω_s) = 0.55π radians
 - Pass band ripple (δ_p) = 0.11
 - And stop band ripple (δ_s) = 0.21

Consider sampling frequency 0.5Hz

Also, convert the obtained digital low-pass filter to high-pass filter with new pass band frequency (ω'_p) = 0.45π using digital domain transformation.
10. Why do we need FFT? Find 8-point DFT of sequence $x[n] = \{1, 1, 2, 2, 1, 1, 2, 1\}$ using Decimation in frequency FFT (DIFFT) algorithm. [2+7]
11. Find $x_3[n]$ if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k) X_2(k)$ where $X_1(k)$ and $X_2(k)$ are 4-point DFT of $x_1[n] = \{1, 2, -2\}$ and $x_2[n] = \{1, 2, 3, -1\}$ respectively. [6]